Rogue Quantum Shooter

1. Title, developers, description

- Game title: Rogue Quantum Shooter
- Developers:
    - Maksims Mirgaļejevs (mm23175),
    - Mareks Vasiļevskis (mv23041)
- Genre: Top-down 2D roguelike shooter
- Concept: Survive a set amount of time in a large arena while fighting aliens and monsters

2. Implemented features & current gameplay

- Top-down player movement (WASD/arrows) with 8-directional idle/run animations
- Automatic shooting (fire rate, bullet projectiles, muzzle particles)
- Camera following the player
- Basic chasing enemies that hunt the player
- They spawn randomly around the player. spawn_interval is 2.5sec. max_enemies currently alive is 20,
- Bullet collision detection with enemy; Enemy death on their hp=<0
- X Gate ressurection/suicide aid kit and HUD text showing left X Gate charges
- Entangle Bullet Skill - either kills enemy and collapses player to ALIVE from ANY state (including the DEAD state - its called Quantum Resurrection mechanic), or kills player and enemy stays alive, HUD text with timer attached.
- **Quantum state system** wired into the player:
- 4 states: |1⟩ (alive), |+⟩ (doomed phase), |−⟩ (safe phase), |0⟩ (dead)
- Full hit cycle: first hit → superposition, subsequent hits are death rolls with 50/50 chances
- HUD showing the current quantum state, game over screen on death, entangle bullet results
- Automated test suite (GUT): unit tests for the quantum core + integration/scene tests (enemy behavior, player wiring, HUD)
- Playable Windows prototype build published as a GitHub release

3. Quantum concepts & mechanics

Implementation:

- The player's state is a simulated qubit - two complex amplitudes (|0⟩ = dead, |1⟩ = alive) tracked as 2D vectors.
- Gates are actual unitary operations on the amplitudes. the visible state is derived from |amplitude|² and the relative phase.

- Superposition - the first enemy hit applies the Hadamard gate $H|1\rangle$, putting the character into superposition: neither alive nor dead, displayed on the HUD.
- Hidden phase - $|+\rangle$ and $|-\rangle$ look identical to direct observation (both 50/50), but the hidden relative phase decides the outcome at the Gate.
- Measurements - every hit taken in superposition is a measurement attempt: base death chance 50% in $|-\rangle$, 80% in $|+\rangle$, scaled by Luck (up to ×0.2 reduction at max Luck).
- Z gate (phase flip) - surviving a hit in $|-\rangle$ applies Z: $|-\rangle \rightarrow |+\rangle$. The "safe" phase degrades into the "doomed" phase.
- Interference & the Gate - (design) the end-of-level Gate applies H and measures: $H|-\rangle = |1\rangle$ (passes, deterministic), $H|+\rangle = |0\rangle$ (dies). Items can apply Z flips; two identical flips cancel out.
- Rz(θ) rotation / Luck - (design) Luck is a rotation around the Z axis; accumulating π flips $|+\rangle$ back to $|-\rangle$. Earned by killing enemies quickly and in close combat — pushes the player to play aggressively exactly when most vulnerable.
- Entanglement - (design) Entangle Bullet skill: link player ←→ enemy; measuring the enemy into $|0\rangle$ leaves the player $|1\rangle$. High Luck favors the player; failure collapses the player to $|0\rangle$.
- X Gate applies flips 0 -> 1; 1 -> 0. Meaning it may ressurect the player character from the dead state, or actually kill the player. If they are not careful enough.

Gameplay effect:

- Combat is a risk engine the player must read their state, farm Luck aggressively in $|+\rangle$, and manage phase and item order instead of just dodging enemies and bullets.

4. Not Yet Implemented / Plans for the Final Version

  - Z phase flip on surviving a hit in $|-\rangle$ ($|-\rangle \rightarrow |+\rangle$)
- End-of-level Gate with H + measurement and interference-based pass/fail
- Luck economy: earning Luck from fast/close kills, Rz(π) accumulation flipping $|+\rangle \rightarrow |-\rangle$
- Items & upgrades: weapons, armor, quantum items (Z flips, combos that cancel), 3–5 card picks on level up
- Run structure: timer-based arena levels, final Gate spawn, progression through multiple levels
- Second play mode idea: $|0\rangle$ as a playable "injured" state with HP split across states (50/100/150 HP) (Optional)
- Quantum enemies - superposition enemies that must be measured to be destroyed for good (Optional, health bar may be more entertaining)
- Polish: more enemy types, boss, music, SFX

5. AI Usage

AI coding assistant Command Code / deepseek v4.1 flash was used for:

- Learning through gdscript and other aspects of godot engine

PixelLab for main character asset generation (do not really like the design we will reinvestigate this in future)

AI ChatGPT:

- Godot engine structure and scripts; debugging

6. Contributions (Group Project)

- Mareks Vasiļevskis (mv23041): core gameplay programming (movement, shooting, enemy), quantum state implementation, HUD, test suite, engine setup, build & release pipeline.
- Maksims Mirgaļejevs (mm23175): game concept and story, quantum mechanics design (states, gates, measurements, interference, entanglement), design documentation, testing documentation, gameplay programming (bullet collision detection with enemies; enemy spawning around player (in the manner of roguelike games); X Gate item and its HUD; Entangle Bullet and its HUD); build & release pipeline.